

Growth-Share Matrix as a Tool for Portfolio Planning: Evidence from the Indian Telecommunication Services Industry

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Portfolio analysis is used for making future strategic investment decisions for diversified streams of business for a firm. This analysis will help the corporate firms to do effective resource allocation for securing future competitive positions. The sample units comprises 17 telecommunication firms which are providing either all or a combination of seven selected services for the purpose of study, namely, wireless GSM, wireless CDMA, wireline, internet inclusive of broadband services, wireless and wireline broadband services, Village Public Telephone (VPT) and Public Call Offices (PCO). The growth-share matrix proposed by the Boston Consulting Group (BCG) as a tool of portfolio planning has been developed for each of the selected services with the bubble size in the matrix reflecting the subscriber base in millions for the respective service line by March 2009. The wireless GSM and CDMA services are currently a question mark, wireline telecommunication services are located in the dog quadrant, the internet services and broadband services standalone are positioned in the question mark, the VPT Booth service is a question mark and the PCO segment is positioned in the dog quadrant for majority of the selected sample units. The results are used to suggest portfolio planning strategies for the sample units.

Introduction

The major objectives of the strategic planning process are of constituting Strategic Business Units (SBUs) in accordance with the corporate mission, planning new businesses and assigning resources to these SBUs (Kotler, 1994). Effective planning would design a perfect balance between the resources and the strategic intent (vision, program, procedures) amidst the environmental uncertainty (Drucker, 1990). Any organization having a portfolio of products or brands requires strategic planning for prudent management of cash with the tradeoff between cash generation and cash usage (McDonald, 1989). The portfolio model is considered very useful for internal analysis and development of strategic marketing options with respect to resource allocation decisions (Aaker, 1998). Portfolio matrix can be initiated by using a pair of variables strategic to the organization and including more than one SBU in the same analysis. Business portfolios can be assessed at the level of markets, segments within markets and distribution outlets for insights into strategic planning (Dibb, 1995).

Relevance of the Portfolio Matrix

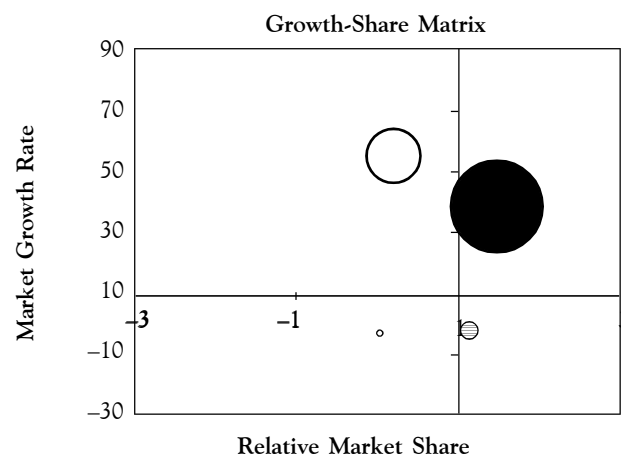
The Boston Consulting Group (BCG) developed a portfolio matrix for more strategic allocation of funds based on cash returns (Tiles, 1966; and Henderson, 1970) and it is a two-dimensional classification of products according to relative market share (cash usage)

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and market growth rate (cash generation) (Dibb, 1995) (Figure 1). This BCG approach has been widely described in the journals for strategic planning (Day, 1977; Hedley, 1977; Wensley, 1981; and Wind and Mahajan, 1981). The matrix helps in development of plans which reflects the need of each business unit as well as business as a whole (Bell, 1982) and also compares the strategic positions of company's diversified business portfolio of investments (Thompson and Strickland, 1993). The strategic planning tool would eventually integrate the strategy across all business units (Day, 1975; and Derek and John, 1979). Although the matrix has limitations (Hofer and Schendel, 1978), it is widely cited in academic and popular discussions of strategy. Other more refined portfolio matrices, such as those used by Arthur D Little (Patel and Younger, 1978), Royal Dutch Shell (Robinson *et al.*, 1978), and General Electric (Taylor, 1976) generally are consistent with the BCG framework, because they all incorporate an 'attractiveness' dimension (broader than, but encompassing, life cycle) and a 'competitive position' dimension (broader than, but encompassing market share).

Figure 1: Nomenclature of the Four Quadrants of the Growth-Share Matrix



Note: The bubble size reflects the relative size of the product asset placed in each of the quadrants.

Market Growth Rate

The vertical dimension of the matrix is 'market growth rate'. For most products, growth rates closely correspond with certain stages of the life cycle. The conceptual distinction is that each stage typically is attributed with characteristics in addition to growth rate, for example, customer adoption rates and the nature of competition. It is an important determinant of resource allocation and it has been claimed that volume gains above market growth rate cut off would be easier to achieve in high growth markets because competition will be less aggressive in response to the implied loss of additional volume rather than the direct reduction in volume that would occur in the case of low growth markets (Wensley, 1982).

Relative Market Share

The horizontal dimension is 'relative market share' which is a key factor affecting the performance of business units. It also points towards the cash position being achieved by the product which may be either high or low generation of cash balances and surpluses. The weight of evidence indicates that high share businesses have significantly higher earnings than do low share businesses (Chevalier, 1972; and Schoeffler *et al.*, 1974). It is possible for businesses to have (or buy) more market share than is optimal (Fruhan, 1972). The market share is a surrogate measure of a firm's position on the experience curve (Bruce, 1968) as a firm (or SBU) with the largest share of a market will be the most profitable at the prevailing price level because it will have the lowest unit costs and highest margins. Less experienced competitors will have lower market shares and, therefore, higher unit costs and lower margins. Consequently, the maxim in this instance is that margins, and generated cash flow, increase with relative market share. (Douglas and Christopher, 1985). This model can be further developed by dilating the point representing a product's position in terms of the given dimensions, to form a circle, the area (or diameter) of which is proportional to sales.

The Quadrants of the Portfolio Matrix

There is not unanimity as to what the dividing points on the two dimensions of the matrix should be. BCG favors 10% real market growth as the point of distinction between high and low market growth. The group views market share in terms of the ratio of the share held by the business relative to the share held by its leading competitor. A ratio of 1.0, indicating highest market share, is commonly considered the dividing line between high and low share. BCG also states that a ratio of 1.5 is necessary to claim and exploit true dominance in a market (Hedley, 1977). This categorizes the firm's products into four groups in terms of relative market share and growth rate and charts them in a two-dimensional, four-quadrant matrix. As illustrated in Figure 1, products with a large market share but slow growth are called 'cash cows'. Those with large share and high growth are 'stars', while those with fast growth but relatively small market share are classified as 'question marks'. Finally, there are the 'dogs', which have slow growth and small shares. The portfolio composition is a function of the balance between cash flows. High market growth segments typically require cash infusions. Low growth segments can be expected to generate excess cash (Table 1). Thus BCG provides performance/potential-based names

Table 1: Strategic Attributes of the BCG Quadrants			
BCG Classifications	Cash Position	Life Cycle Stage	Risk Areas
Star	Generation of large cash balances	Growth with consolidation of competitive position of the product	Cuts in investments to gain short-term returns turns Stars into Dogs
Cash Cow	Generation of large cash surpluses	Maturity with slower market growth rate of the product sales	Managing cash cow only for cash. Product substitution in the market

Table 1 (Cont.)

BCG Classifications	Cash Position	Life Cycle Stage	Risk Areas
Question Mark	Cash generation is Low	Market growth rate is high but firm's sales growth slow	No Investment for gaining competitive position turns Question marks into Dogs
Dogs	Cash usage	Mature Markets and low product sales growth rate	Little potential for profitability

for the SBUs according to the quadrant into which they fall, along with possible strategies to consider for SBUs positioned in each quadrant (Gomes and Knowles, 1997).

Research Methodology

The sample units comprises 17 telecommunication firms which are providing either all or a combination of seven selected services for the purpose of study, namely, wireless GSM, wireless CDMA, wireline, internet inclusive of broadband services, wireless and wireline broadband services, Village Public Telephone (VPT) and Public Call Offices (PCO). The growth-share matrices have been developed for each of the selected service with the bubble size in the matrix reflecting the subscriber base in millions for the respective service line of each sample unit as on March 2009 (Table 2). The market growth rate of the service is taken as the vertical dimension (Table 3) while the horizontal dimension is taken as the relative market share of each of the sample unit providing the service (Table 4). This relative market share is the ratio of the operator's number of subscribers and that of the market leader for the selected service. In case of the leading operator, the ratio is calculated by using

**Table 2: Asset Size or Subscribers in Millions
by March, 2009 of the Sample Units**

Players	Wireless GSM	Wireless CDMA	Wireline	Internet (Including Broad-band)	Broad Band	VPT	PCO
Bharti Airtel Ltd.	93.92	–	2.73	1.080	0.81	–	0.17
Vodafone Essar Mobile Services Limited	68.77	–	–	0.018	–	–	–
Bharat Sanchar Nigam Ltd.	15.71	5.44	29.34	7.250	3.50	0.550	1.90
Idea	38.89	–	–	–	–	–	–
Reliance Communications Infrastructure Limited	20.02	52.65	1.11	0.940	0.11	0.004	2.14
Aircel	18.48	–	–	–	–	–	–
Mahanagar Telephone Ltd.	4.18	0.30	3.57	2.120	0.70	–	0.22
Spice	4.13	–	–	–	–	–	–
BPL	2.16	–	–	–	–	–	–

Table 2 (Cont.)

Players	Wireless GSM	Wireless CDMA	Wireline	Internet [Including Broad- band]	Broad Band	VPT	PCO
Tata Teleservices	–	35.12	0.92	0.37	0.22	0.004	1.74
Shyam Telelink	–	0.60	0.13	0.033	–	0.003	0.04
HFCL Infotel Ltd.	–	0.39	0.16	0.07	0.06	0.0002	0.03
Sify Technologies	–	–	–	0.41	0.09	–	–
Hathway Cable & Datacom Pvt. Ltd.	–	–	–	0.34	0.27	–	–
Data Infosys Ltd.	–	–	–	0.25	–	–	–
You Telecom Pvt. Limited	–	–	–	0.20	0.15	–	–

**Table 3: Industry Data for Services Mix,
Market Size and Growth Rate for the Period March 2008-March 2009**

S. No.	Telecommunication Service Segments in Study	Market Size in India (March 2009) in Millions of Subscribers	Annual Market Growth Rate in 2009 (% Growth in Subscriber Base Over March 2008)
1.	Wireless GSM	297.260	54.26
2.	Wireless CDMA	94.500	38.21
3.	Wireline	37.960	–3.70
4.	Internet including broadband	13.540	22.09
5.	Broadband connections	6.220	60.72
6.	Village Public Telephone (VPT)	0.561	17.85
7.	Public Call Office (PCO)	0.619	–3.06

Table 4: Relative Market Shares of Sample Units

Sample Units	Wireless GSM	Wireless CDMA	Wireline	Internet (Including Broad- band)	Broad- band	VPT	PCO
Bharti Airtel Ltd. (BAL)	1.36	–	0.09	0.15	0.23	–	0.87
Vodafone Essar Mobile Services Limited (VEMSL)	0.73	–	–	–	–	–	–
BSNL	0.50	0.1	8.21	3.42	4.36	133.490	0.88
Idea	0.41	–	–	–	–	–	–
Reliance Communications Infrastructure Limited (R)	0.21	1.5	0.037	0.13	0.03	0.007	1.14

Table 4 (Cont.)

Sample Units	Wireless GSM	Wireless CDMA	Wireline	Internet (Including Broad-band)	Broad-band	VPT	PCO
Aircel	0.20	–	–	–	–	–	–
Mahanagar Telephone Ltd. (MTNL)	0.04	0.006	0.1210	0.290	0.20		0.10
Spice	0.04	–	–	–	–	–	–
BPL	0.02	–	–	–	–	–	–
Tata Teleservices (TTSL)	–	0.670	0.0310	0.050	0.06	0.007	0.81
Shyam Telelink (STS)	–	0.010	0.0038	0.005	–	0.005	0.02
HFCL Infotel Ltd. (HFCL)	–	0.010	0.0051	0.010	0.02	4.010	0.01
Sify Technologies (SIFY)	–	–	–	0.060	0.03	–	–
Hathway Cable & Datacom Pvt. Ltd. (HCDP)	–	–	–	0.050	0.07	–	–
Data Infosys Ltd. (DIL)	–	–	–	0.030	–	–	–
You Telecom Pvt. Limited (YTPL)	–	–	–	0.030	0.04	–	–

the data of the next best competitor as the denominator. The present study relies on the 10% relative market growth rate and 1.0 relative market share as the dividing points in the matrix. Data has been collected from the TRAI report published in March 2009.

Results

The wireless GSM service is currently a 'question mark' for all the sample units except for Bharti Airtel Limited wherein the service is positioned in the 'star' quadrant (Figure 2). The CDMA segment of wireless services is also a 'question mark' for most of the players (Figure 3) except for Reliance Communication Infrastructure Ltd. (star) and HFCL (dog). The sample units offering wireline telecommunication services (Figure 4) are located in the dog quadrant except for BSNL (cash cow). The internet services (Figure 5) and broadband services standalone (Figure 5) showed similar results and are positioned in the 'question mark' quadrant except for BSNL wherein they are placed in the 'star' quadrant. The VPT Booth service (Figure 7) is a 'star' for BSNL and HFCL and a 'question mark' for the other selected sample units. The PCO segment (Figure 8) is positioned in the 'dog' quadrant for all the players except for BSNL (cash cow). The relative asset size of the state-owned operator BSNL is quite big as compared to other players in all the segments except in wireless CDMA services. It is observed BSNL has a monopoly in terms of asset size, i.e., number of subscribers in the segments of wireline, internet including broadband and broadband measured separately through the bubble size. The wireless GSM services are highly competitive in terms of asset size with a consolidated market growth rate of 54.26%.

The number of 'question marks' is relatively higher for the portfolio of services studied in the sample units. Eight sample firms have only 'question marks' in their portfolio of services indicating low market shares in high growth segments while 4 sample units have either 'question marks' or 'dogs' as their product offerings indicating low market share irrespective of growth rate of segments (Table 5). Vodaphone Essar Mobile Services Limited is the only sample unit having only 'star' and 'question marks' in their portfolio thus operating only in the high growth market segments. State-owned BSNL has an

Figure 2: Relative Positioning of Operators Providing Wireless GSM Services in Growth Share Quadrants

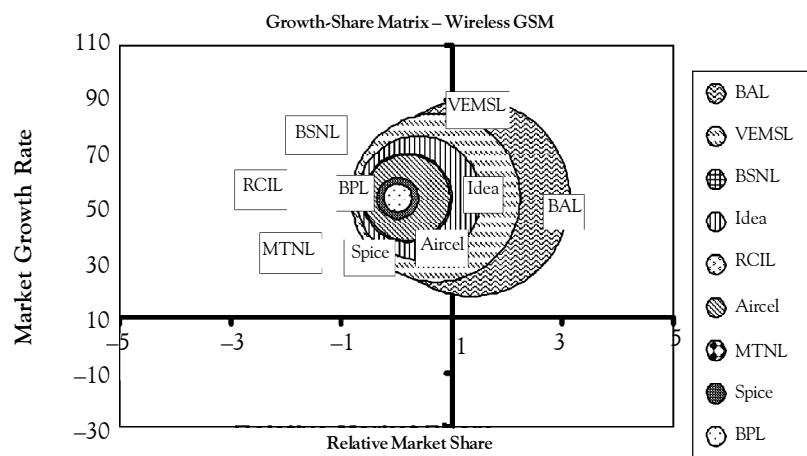


Figure 3: Relative Positioning of Operators Providing Wireless CDMA Services in Growth Share Quadrants

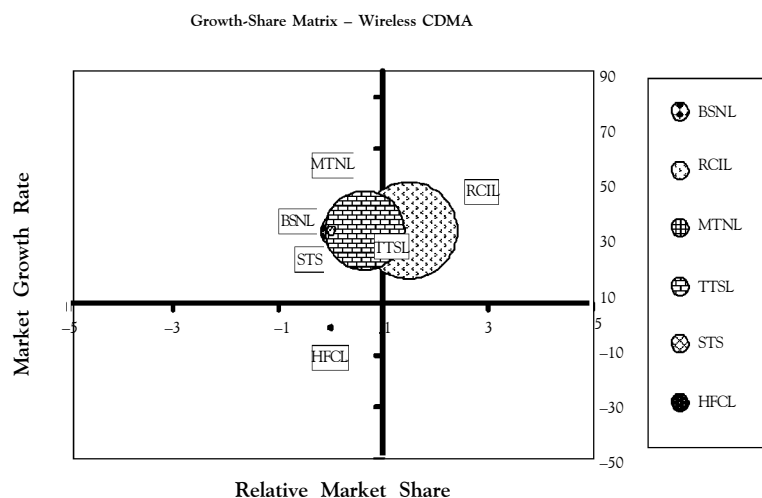


Figure 4: Relative Positioning of Operators Providing Wireline Services in Growth-Share Quadrants

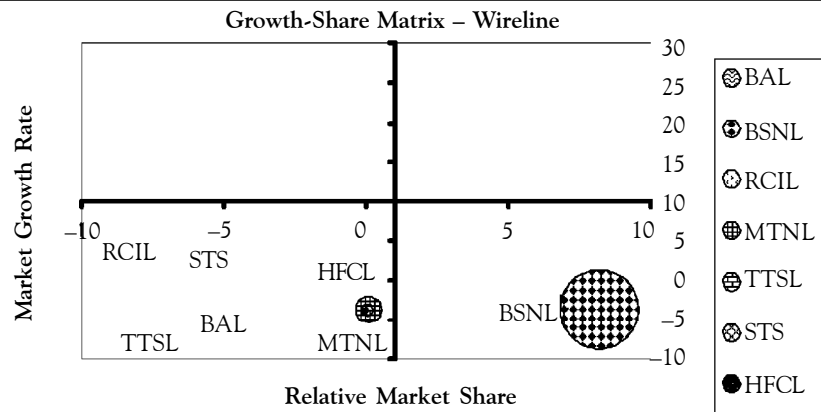


Figure 5: Relative Positioning of Operators Providing Internet Services in the Growth-Share Quadrants

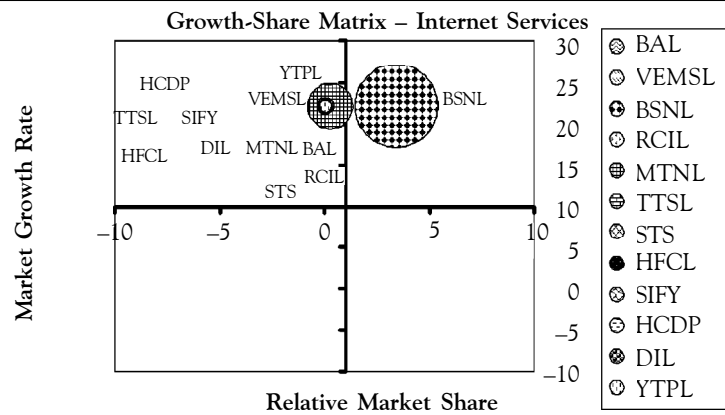


Figure 6: Relative Positioning of Operators Providing Broadband in the Growth-Share Quadrants

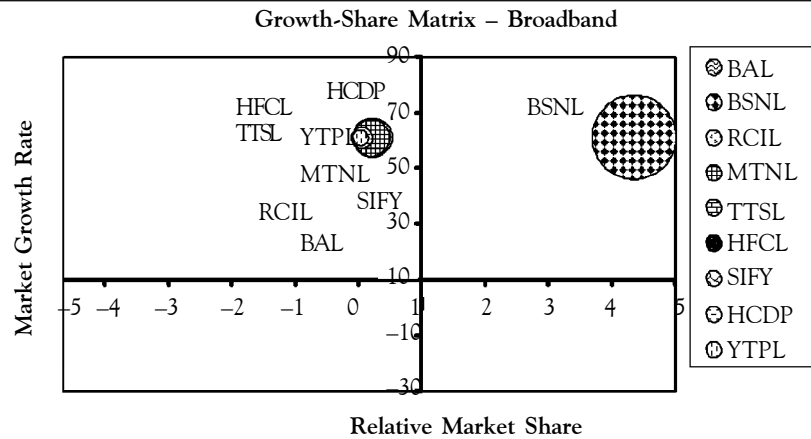


Figure 7: Relative Positioning of Operators Providing VPT in the Growth-Share Quadrants

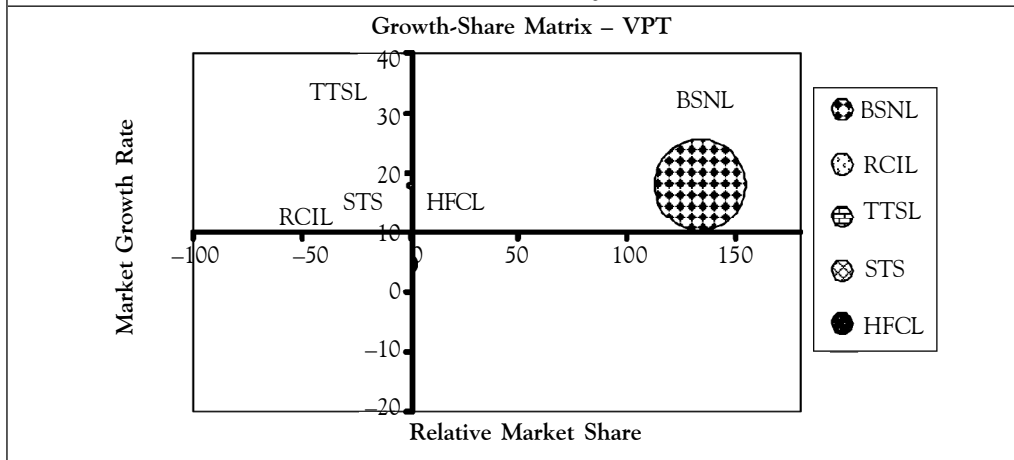
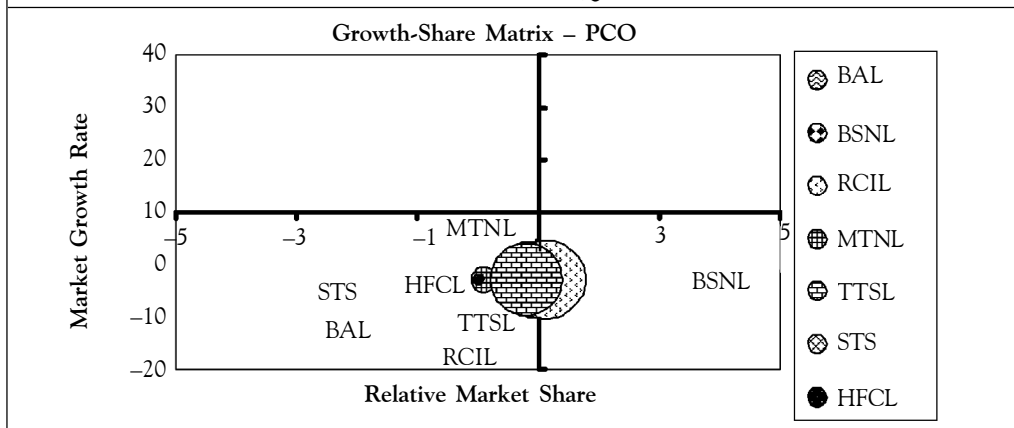


Figure 8: Relative Positioning of Operators Providing PCO in the Growth-Share Quadrants



attractive portfolio comprising of 'stars', 'question marks', 'cash cow' except one 'dog'. Reliance Communications Infrastructure Limited has three 'question marks' and one each of 'star', 'cash cow' and 'dog' in its portfolio with an immediate need to consolidate its market share in high growth segments.

Discussion and Conclusion

Modeling an Ideal Portfolio: Relatively high number of stars put a strain on the cash resources, relatively high number of cash cows result in excessive cash flow but future growth is uncertain and too many dogs and question marks generate inadequate cash flows (Douglas and Christopher, 1985). The BCG form of analysis has its roots in the concept of a competitive market place (Wensley, 1982). So, the portfolio matrix has to result in competitive resource allocation decisions (Table 1). The wireless GSM and CDMA, internet and broadband and VPT services are generating cash balances and need big level

of investments to consolidate market position in the growth phase of the life cycle as there are substantial cash costs of funding projects in high growth markets (Wensley, 1981). However, some of the operators providing the above services are positioned in the 'question marks' quadrant and may face the risk of cuts in investments for short-term returns owing to low market share. Wind *et al.* (1983) have contested that any innovative product/service would also be positioned as 'question mark' with huge growth potential but zero market share and may mislead the planning process. The remaining portfolio of services, namely, wireline and PCO are mostly generating cash surpluses in the short-term and are in the maturity phase of the life cycle. The recommended strategy would be to look for product substitution involving future investments into new or other high growth businesses. However, the segment has reached the decline stage for one of the sample unit

Table 5: Portfolio Composition of the Sample Units

Sample Units	Stars	Question Mark	Cash Cows	Dogs
Bharti Airtel Ltd.	2	2	–	1
Vodafone Essar Mobile Services Limited	1	1	–	
Bharat Sanchar Nigam Ltd.	2	2	1	1
Idea	–	1	–	–
Reliance Communications Infrastructure Limited (RCIL)	1	3	1	1
Aircel	–	1	–	–
Mahanagar Telephone Ltd. (MTNL)	–	4	–	2
Spice	–	1	–	–
BPL	–	1	–	–
Tata Teleservices (TTSL)	–	3	–	1
Shyam Telelink (STS)	–	3	–	2
HFCL Infotel Ltd. (HFCL)	–	3	–	2
Sify Technologies (SIFY)	–	2	–	–
Hathway Cable & Datacom Pvt. Ltd. (HCDP)	–	2	–	–
Data Infosys Ltd. (DIL)	–	1	–	–
You Telecom Pvt. Limited (YTPL)	–	2	–	–

and requires divestment and exit strategies. Sometimes though the interdependence of business units can be a barrier to exit. Resource allocation decisions can be validated through review BCG matrices after one business cycle.

Limitations of the Study: Proctor and Hassard (1990) have pointed out that there might be other important attributes, namely, environmental shift besides market share and growth rate for considerations in the portfolio matrix. Wensley (1981) has pointed out the inadequacy of application of this matrix for SBUs in niche service or product segments.

Also, the relationship of the market share with other objectives of the organization is also unclear (Kerin *et al.*, 1990) especially, in the case of not for profit SBUs. However, the opportunity for contribution in this area is considerable, both because portfolio methods are in high use (Slater and Zwirlein, 1992), and because they are often misused, or not used to their fullest potential (Gomes and Knowles, 1997). ☛

References

1. Aaker D A (1998), *Strategic Market Management*, 5th Edition, John Wiley & Sons, New York.
2. Bell M L (1982), "Advertising Strategy and Strategic Market Planning", in A D Fletcher (Ed.), *Proceedings of the 1982 Conference of the American Academy of Advertising*, pp. 7-11, University of Tennessee, Knoxville, TN.
3. Bruce D Henderson (1968), "Profit Centers and Decentralized Management", W Stern Carl and Michael S Deimler (Eds.), *The Boston Consulting Group on Strategy*, pp. 282-285, John Wiley & Sons, Reprinted in 2006.
4. Chevalier M (1972), "The Strategy Spectre Behind your Market Share", *European Business*, Vol. 34, pp. 63-72.
5. Day G S (1975), "A Strategic Perspective on Product Planning", *Journal of Contemporary Business*, Vol. 4, pp. 1-34.
6. Day G S (1977), "Diagnosing the Product Portfolio", *The Journal of Marketing*, Vol. 41, No. 2, pp. 29-38.
7. Derek F Abell and John S Hammond (1979), *Strategic Market Planning: Problems and Analytical Approaches*, Englewood Cliffs, Prentice-Hall, NJ.
8. Dibb Sally (1995), "Developing a Decision Tool for Identifying Operational and Attractive Segments", *Journal of Strategic Marketing*, Vol. 3, pp. 1-15.
9. Douglas T Brownlie and Christopher K Bart (1985), "Products and Strategies", *Management Bibliographies & Reviews S*, ISBN-13: 978-0861762293, MCB Univ. Press, UK.
10. Drucker Peter F (1990), *Managing the Nonprofit Organization: Principles and Practices*, Harper Business, New York.
11. Fruhan W E Jr. (1972), "Pyrrhic Victories in Fights for Market Share", *Harvard Business Review*, Vol. 50, No. 5.
12. Gomes Roger and Knowles Patricia (1997), "The Customer Value/Mission Matrix and an Application Process for all Portfolio Planning", *The Journal of Marketing Management*, Vol. 7, No. 2, pp. 34-43.
13. Hedley B (1977), "Strategy and the Business Portfolio", *Long Range Planning*, Vol. 10, No. 1, pp. 9-15.
14. Henderson B (1970), "The Product Portfolio", *BCG Perspectives*, p. 66.

15. Henderson (1979), *B D Henderson on Corporate Strategy*, Abt Books, Cambridge, Mass.
16. Hofer C W and Schendel (1978), *D Strategy Formulation: Anatyticat Concepts*, West, St. Paul.
17. Kerin R A, Mahajan V and Varadarajan P R (1990), *Contemporary Perspectives on Strategic Market Planning*, Allyn and Bacon, Boston, MA.
18. Kotler Philip (1994), *Marketing Management: Analysis, Planning, Implementation, and Control*, 8th Edition, Englewood Cliffs, Prentice-Hall, NJ.
19. McDonald M H B (1989), *Marketing Plans. How to Prepare Them: How to Use Them*, Butterworth-Heinmann, Oxford.
20. Patel P and Younger M A (1978), "Frame of Reference for Strategy Development", *Long Range Planning*, Vol. 11, pp. 6-12.
21. Proctor R A and Hassard J S (1990), "Towards a New Model for Product Portfolio Analysis", *Management Decision*, Vol. 28, May, pp. 14-17.
22. Robinson S J Q, Hickens R E and Wade D (1978), "The Directional Policy Matrix—Tool for Strategic Plannings", *Long Range Planning*, Vol. 10, pp. 17-27.
23. Schoeffler S, Buzzell R and Heany D (1974) "Impact of Strategic Planning on Profit Performance", *Harvard Business Review*, Vol. 52, No. 1, pp. 137-145.
24. Slater Stanley F and Thomas I Zwirlein (1992), "Shareholder Value and Investment Strategy Using the General Portfolio Model", *Journal of Management*, Vol. 18, No. 4, pp. 717-732.
25. Taylor B (1976), "Managing the Process of Corporate Development", *Long-Range Planning*, Vol. 9, pp. 81-100.
26. Tiles S (1966), "Strategies for Allocating Funds", *Harvard Business Review*, Vol. 44, pp. 72-80.
27. Thompson Jr. A A and Strickland III A J (1993), *Strategic Management: Concepts & Cases*, 7th Edition, Irwin, Homewood, IL.
28. Wind Y and Mahajan V J (1981), "Designing Product & Business Portfolios", *Harvard Business Review*, Vol. 59, pp. 155-165
29. Wind Y, Mahajan V and Swire D J (1983), "An Empirical Comparison of Standardized Portfolio Models", *Journal of Marketing*, Vol. 47, Spring, pp. 89-99.
30. Wensley R (1981), "Strategic Marketing; Betas, Boxes or Basics", *Journal of Marketing*, Summer, pp. 173-183.
31. Wensley R (1982), "PIMS and BCG: New Horizons or False Dawn?" *Strategic Management Journal*, Vol. 3, pp. 147-158.

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